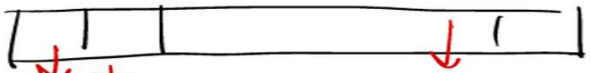
Introduction to Probability and Statistics

(EE 354 / CE 361)

Counting: Part 2

Recap

(1) n unique objects, # of ways to arrange



$n \times (n-1) \times \cdots \times 2 \times 1 = n!$

(2) n – objects and I want to find # of subsets

$= 2^n = \underbrace{2 \cdot 2 \cdot \cdots \cdot 2}_{n \text{ times}}$

(3) n – objects, k are selected & then arrange them — k-Permutations

$$\frac{n!}{(n-k)!} = {}^nP_k$$

(4) n – object, k are selected (arrangement not done)

$${}^nC_k = \frac{n!}{k!(n-k)!} = \binom{n}{k}$$

Today's Agenda

- Partitions
- Independent Bernoulli Trials

Partitions

We are given an n-element set and nonnegative integers $n_1, n_2, \ldots, n_r$, whose sum is equal to n. We consider partitions of the set into r disjoint subsets. with the ith subset containing exactly n_i elements. Let us count in how many ways this can be done.

We form the subsets one at a time. We have $\binom{n}{n_1}$ ways of forming the first subset. Having formed the first subset, we are left with $n - n_1$ elements. We need to choose n_2 of them in order to form the second subset, and we have $\binom{n-n_1}{n_2}$ choices, etc. Using the Counting Principle for this r-stage process, the total number of choices is

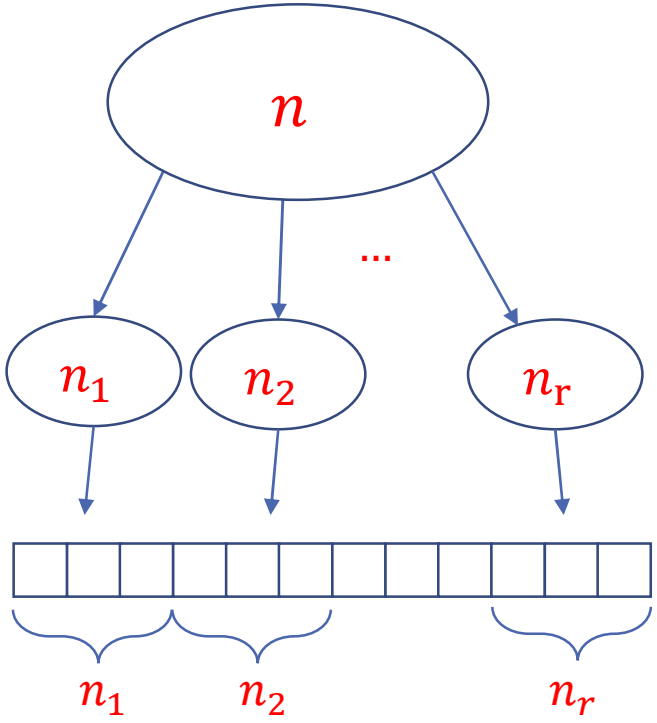
$$\binom{n}{n_1}\binom{n-n_1}{n_2}\binom{n-n_1-n_2}{n_3}\cdots\binom{n-n_1-\cdots-n_{r-1}}{n_r},$$

which is equal to

$$\frac{n!}{n_1!\,(n-n_1)!}\cdot\frac{(n-n_1)!}{n_2!\,(n-n_1-n_2)!}\cdots\frac{(n-n_1-\cdots-n_{r-1})!}{(n-n_1-\cdots-n_{r-1}-n_r)!\,n_r!}.$$

We note that several terms cancel and we are left with

$$\frac{n!}{n_1!\,n_2!\cdots n_r!}.$$



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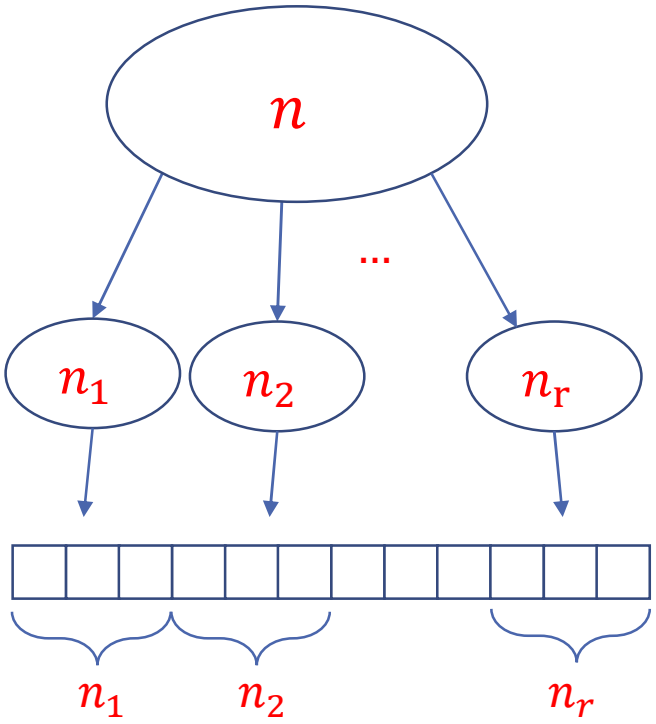
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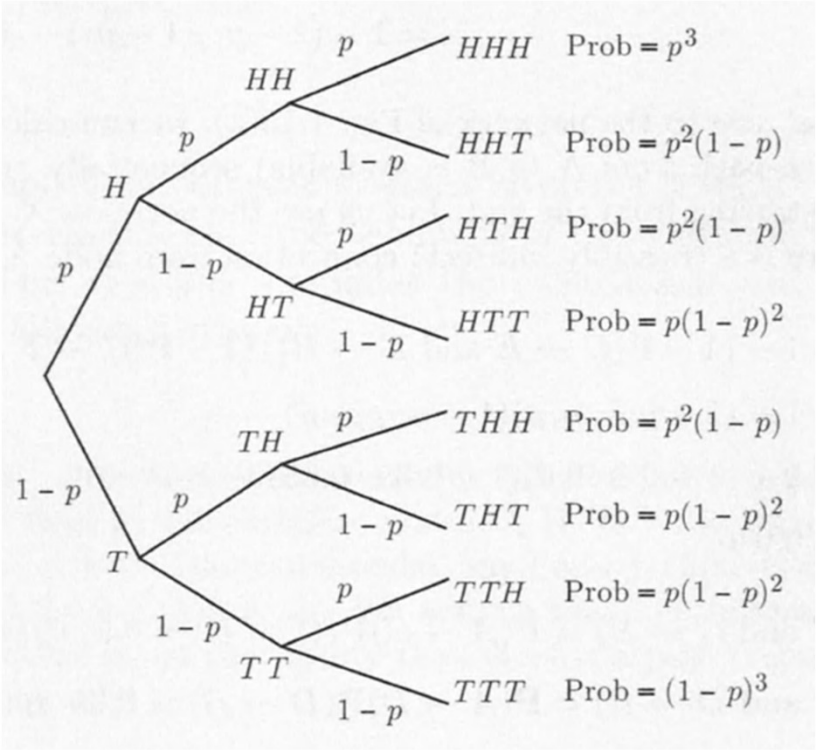
$$\frac{n!}{n_1!\,n_2!\cdots n_r!}.$$



$$\frac{n!}{n_1!\,n_2!\cdots n_r!} = \textbf{multinomial coefficient} = \binom{n}{n_1, n_2, \ldots, n_r}$$

Independent Bernoulli Trials

- Independent Trials
  - If an experiment involves a sequence of independent but identical stages, we say that we have a sequence of independent trials.
- Independent Bernoulli Trials
  - In the special case of independent trials where there are only two possible results at each stage, we say that we have a sequence of independent Bernoulli trials.
- Example
  - Consider the experiment that consists of 3 independent tosses of a coin.



Independent Bernoulli Trials

- Consider an experiment that consists of n independent tosses of a coin, in which the probability of heads is p

Let us now consider the probability

$$p(k) = \mathbf{P}(k \text{ heads come up in an } n\text{-toss sequence})$$

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$\mathbf{P}(H\ T\ T\ H\ H\ H) = \boldsymbol{p.(1-p).(1-p).p.p.p}$

$\mathbf{P}(particular\ sequence) = \boldsymbol{p^{\#\ of\ H}(1-p)^{\#\ of\ T}}$

$\mathbf{P}(particular\ k-head\ sequence) = \boldsymbol{p^k(1-p)^{n-k}}$

$p(k) = \mathbf{P}(k \text{ heads come up in an } n - \text{toss sequence}) = \boldsymbol{\binom{n}{k} p^k(1-p)^{n-k}}$

The numbers $\binom{n}{k}$ (read as "n choose k") are known as the **binomial coefficients**, while the probabilities $p(k)$ are known as the **binomial probabilities**.

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Note that the binomial probabilities $p(k)$ must add to 1, thus showing the **binomial formula**

$$\sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = 1.$$

Binomial Probabilities: Example

Example 1.25. Grade of Service. An internet service provider has installed c modems to serve the needs of a population of n dialup customers. It is estimated that at a given time, each customer will need a connection with probability p, independent of the others. What is the probability that there are more customers needing a connection than there are modems?

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Here we are interested in the probability that more than c customers simultaneously need a connection. It is equal to

$$\sum_{k=c+1}^{n} p(k),$$

where

$$p(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

are the binomial probabilities. For instance, if $n = 100$, $p = 0.1$, and $c = 15$, the probability of interest turns out to be 0.0399.

This example is typical of problems of sizing a facility to serve the needs of a homogeneous population, consisting of independently acting customers. The problem is to select the facility size to guarantee a certain probability (sometimes called **grade of service**) that no user is left unserved.